



University
of Basel

Foundations of Distributed Systems

Fall Semester 2026

Prof. Dr. Florina Ciorba

Prof. Dr. Heiko Schuldt

Prof. Dr. Christian Tschudin

Prof. Dr. Isabel Wagner



Example of a Distributed System: Bitcoin

- Bitcoin (BTC) is a distributed system that ...
 - uses Communication
 - manages Data
 - requires Processing
 - has some Privacy features
- BTC is not the core of FDS but highly related. We
 - invite all participants to a critical discussion of Bitcoin
 - hope that his lecture exposes the foundational aspects of this well-known distributed system



Image licensed from Adobe Stock

Disclaimer: after successful completion of this course, you earn ETCS Credit Points, not BTCs.

Dates

Lecture (in presence):

Start: Thursday, September 17, 2026

Dates: – Thursday 10:15 – 12:00 h location: Spiegelgasse 5, 05.002
– Friday 10:15 – 12:00 h location: Spiegelgasse 5, 05.002

Exercises (in presence):

Start: Week #3, i.e., September 29, 2026

Dates: – Tuesday 10:15 – 12:00 h location: Pharmazentrum, Seminar Room U1075

 Thu. 10 - 12 h Fri. 10 - 12 h	Lectures
 Tue. 10 - 12 h	Exercises
 January 20-26, 2027	Oral examination
	8 ETCS

Team

Lecturers:

- Prof. Dr. Florina Ciorba (florina.ciorba@unibas.ch)
- Prof. Dr. Heiko Schuldt (heiko.schuldt@unibas.ch)
- Prof. Dr. Christian Tschudin (christian.tschudin@unibas.ch)
- Prof. Dr. Isabel Wagner (isabel.wagner@unibas.ch)

Assistants:

- Ali Ajorian, MSc (ali.ajorian@unibas.ch)
Exercises 1 and 2 for Chapter 3
- Yiming Wu, MSc (yiming.wu@unibas.ch)
Exercises 3 and 4 for Chapter 4
- Thomas Jakobsche, MSc (thomas.jakobsche@unibas.ch)
Exercises 5 and 6 for Chapter 5
- Valentyna Pavliv, MSc (valentyna.pavliv@unibas.ch)
Exercises 7 and 8 for Chapter 6

Course Objective & Organization

- The goal of the Foundations of Distributed Systems (FDS) course is to **understand the basic principles of distributed and parallel systems**.
- A major question will be how large-scale problems can be **parallelized and distributed** across compute entities (distribution can span the entire spectrum from cores in a multi-core machine up to nodes in a geographically distributed and heterogeneous system) in a scalable, fault-tolerant and privacy-preserving way.
- The topics of the course will be complemented by **theoretical and practical exercises**. After the launch of each exercise, there will be an introduction in the exercise slot; results do not have to be handed in, but it is highly recommended to do (and not to vibe code!) the assignments. A written test will be organised for each exercise to check whether the concepts have been understood.
- In addition, we will organize an embedded **mini seminar with student presentations on selected topics and addressing all parts of this course**.
- The successful completion of exercises (test results) and participation to the mini seminar is prerequisite for the admission to the individual oral examination in January 2027.

Contents of FDS ...

- **Introduction** (Chapter 1)
How do distributed systems look like when seen from a **Computer Network perspective**, from a **High Performance Computing perspective**, and from a **Database perspective**?
What are the commonalities and the differences of these perspectives? How does **Data Privacy** impact the three different perspectives?
- **Applications** (Chapter 2)
Insights into existing selected applications and the type of distributed systems they are running on.

... Contents of FDS ...

Communication (Chapter 3): composing sequential programs leads to problems ...

Our focus is on these problems, not the means to communicate.

- **Concurrency, Leader Election** (Chapter 3.1)
Can we prevent two programs to concurrently declare themselves to be the boss?
- **Byzantine Agreement Problem** (Chapter 3.2)
How to handle adversarial peers? Spoiler: it's about consensus.
- **Group Communication, Remote Procedure Call (RPC)** (Chapter 3.3)
Making life easy for programmers (of distributed systems) ... is not easy
- **Crash Resilience, WAL** (Chapter 3.4)
What MUST be remembered (stored) else the whole distributed system may die?
- **Consensus, Cloud and CRDTs** (Chapter 3.5)
Let's go conflict-free! (Conflict-free Replicated Data Types, CRDTs)

... Contents of FDS ...

- **Data** (Chapter 4)
 - **Self-Organization (P2P)** (Chapter 4.1)
How can systems be managed / data be distributed without central control?
And how can systems organize and stabilize themselves and automatically recover from failures?
 - **Consistency** (Chapter 4.2)
In a distributed system, state and/or data needs to be maintained across several nodes.
How can this be done consistently?
 - **Data Replication** (Chapter 4.3)
What are the basic principles of distributed databases that replicate data across several nodes in a system to achieve better performance and/or a higher degree of availability?

... Contents of FDS ...

- **Computation** (Chapter 5)
 - **Parallelization Abstractions** (Chapter 5.1)

What is parallelism? What types of parallelism do we encounter in hardware and software? What drives the development of parallel applications and the design of parallel and distributed computing systems?
 - **Parallel Architectures and Programming Models** (Chapter 5.2)

What is the architecture of a parallel (super)computer? What role do memory organization and network interconnections play in parallel computing architectures? How do programming models match the parallel architectures?
 - **Parallel Programming** (Chapter 5.3)

How can one write programs for parallel (super)computers, with shared memory and/or distributed memory? How can (inherent) parallelism within an application be best exposed, expressed to exploit the power of HPC?

... Contents of FDS

- **Privacy** (Chapter 6)
 - **Introduction to Privacy** (Chapter 6.1)
What are examples for privacy violations? How can we define privacy? How can we engineer systems to protect privacy? What does data protection regulation say? What is the value of privacy?
 - **Differential Privacy** (Chapter 6.2)
If we have a database with sensitive information, how can we allow queries that give statistically useful responses without leaking sensitive information?
 - **Homomorphic Encryption** (Chapter 6.3)
If we have private data but not enough compute power, how can we outsource our computations while keeping our data private?
 - **Anonymous Communication** (Chapter 6.4)
How can we communicate with others, while ensuring not only confidentiality of messages, but also anonymity of the sender and/or receiver, and while hiding whether we have communicated at all?

Exam and Grading

At the end of the semester, an oral examination (individual, duration: 30 min) will be scheduled.

Dates:

- Wednesday January 20, 2027,
- Thursday January 21, 2027,
- Friday January 22, 2027,
- Monday January 25, 2027, and
- Tuesday January 26, 2027.

There will be four exercise sheets, one per topic, with two assignments each. Each exercise will have 20 points that can be earned from the accompanying EvaExam test per exercise. The practical part of the exercises (that does not have to be handed in) can be done in groups, the EvaExam tests are individual.

Prerequisite for the admission to the oral examinations is the successful completion of the exercises/tests and participation in the mini seminar:

In total, at least 3/4 of the overall points for the exercises must be earned (and at least 50% of each individual exercise).

Course Material

Up-to-date information on the course are available on the course website:

<https://dmi.unibas.ch/de/studium/computer-science-informatik/lehrrangebot-hs26/lecture-foundations-of-distributed-systems/>

Course material, background information, additional material, exercises, etc. will be available on the course website and the ADAM platform of the University of Basel. Access to ADAM will be provided automatically after enrollment to the course (with a 24 hours delay):

<https://adam.unibas.ch/go/crs/2206930>

ADAM will be used for running the exercises; exercise results can be uploaded to ADAM and the system will also be used for grading the exercises.

Course Schedule –1–

The following table depicts the planned course schedule (as of September 2026). This schedule might still be subject to changes in the course of the semester.

Week	Day	Date	Time	Lecturer	Topic	Exercises
1	Thursday	17.09.2026	10 - 12 h	FC, HS, CT, IW	Organization Chapter 1: Introduction (Comm / Data)	
	Friday	18.09.2026	10 - 12 h	FC, HS, CT, IW	Chapter 1: Introduction (Proc / Priv) Chapter 2: Applications (Comm / Proc)	
2	Thursday	24.09.2026	10 - 12 h	FC, HS, CT, IW	Chapter 2: Applications (Data / Priv) Distribution of Mini Seminar Topics	
	Friday	25.09.2026	10 - 12 h	CT	Chapter 3.1: Concurrency, leader election	Ex 1: Causality graph (Git and vector , clocks) RPC
3	Thursday	01.10.2026	10 - 12 h	CT	Chapter 3.2: Byzantine agreement problem	
	Friday	02.10.2026	10 - 12 h	CT	Chapter 3.3: Group communication, RPC	
4	Thursday	08.10.2026	10 - 12 h	CT	Chapter 3.4: Crash resilience, WAL	
	Friday	09.10.2026	10 - 12 h	CT	Chapter 3.5: Consensus, cloud and CRDTs	Ex 2: Consensus
5	Thursday	15.10.2026	10 - 12 h	HS	Chapter 4.1: Self-Organization (P2P)	
	Friday	16.10.2026	10 - 12 h	HS	Chapter 4.1: Self-Organization (P2P)	
6	Thursday	22.10.2026	10 - 12 h	HS	Chapter 4.2: Consistency	Ex. 3: P2P
	Friday	23.10.2026	10 - 12 h	HS	Chapter 4.2: Consistency	

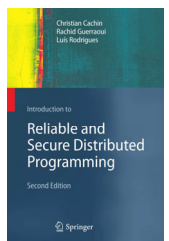
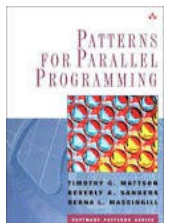
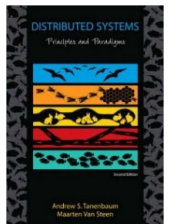
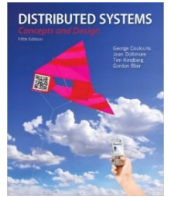
Course Schedule –2–

Week	Day	Date	Time	Lecturer	Topic	Exercises
7	Thursday	29.10.2026	10 - 12 h	FC	Chapter 4.3: Data Replication	Ex. 4: Consistency
	Friday	30.10.2026	10 - 12 h	FC	Chapter 5.1: Parallelization abstractions	
8	Thursday	05.11.2026	10 - 12 h	FC	Chapter 5.2: Parallel Architectures and Programming Models	Ex. 5: OPenMP
	Friday	06.11.2026	10 - 12 h	FC	Chapter 5.2: Parallel Architectures and Programming Models	
9	Thursday	12.11.2026	10 - 12 h	FC	Chapter 5.3: Parallel Programming	Ex. 6: MPI
	Friday	13.11.2026	10 - 12 h	HS	Chapter 5.3: Parallel Programming	
10	Thursday	19.11.2026	10 - 12 h	IW	Chapter 6.1: Introduction to Privacy	
	Friday	20.11.2026	10 - 12 h	IW	Chapter 6.2: Differential Privacy	
11	Thursday	26.11.2026	10 - 12 h	IW	Chapter 6.2: Differential Privacy	Ex. 7: Differential Privacy
	Friday	27.11.2026			<i>No lectures (Dies Academicus)</i>	
12	Thursday	03.12.2026	10 - 12 h	IW	Chapter 6.3: Homomorphic Encryption	
	Friday	04.12.2026	10 - 12 h	IW	Chapter 6.4: Anonymous Communication	
13	Thursday	10.12.2026	10 - 12 h	all	Student Presentations -1-	Ex. 8: Homomorphic Encryption
	Friday	11.12.2026	10 - 12 h	all	Student Presentations -2-	
14	Thursday	17.12.2026	10 - 12 h	all	Overall Discussion	
	Friday	18.12.2026	10 - 12 h	FC, HS CT, IW	Recap, Conclusion, Evaluation, Project Presentations	
	Wednesday	20.01.2027	all day		Oral Examinations	room: tbd
	Thursday	21.01.2027	all day		Oral Examinations	room: tbd
	Friday	22.01.2027	all day		Oral Examinations	room: tbd
	Monday	25.01.2027	all day		Oral Examinations	room: tbd
	Tuesday	26.01.2027	all day		Oral Examinations	room: tbd

Literature

- [CDK⁺ 12] G. Coulouris, J. Dollimore, T. Kindberg, G. Blair: *Distributed Systems: Concept and Design*. Pearson; 5th edition, 2012. ISBN: 978-273-76059-7.
- [TvS 07] A. Tanenbaum, M. van Steen: *Distributed Systems: Principles and Paradigms*. Pearson-Prentice Hall, 2nd edition, 2007. ISBN: 0-13-239227-5.
- [MSM 05] T. Mattson, B. Sanders, B. Massingill: *Patterns for Parallel Programming*. Pearson Education, 2005. ISBN 0-32-122811-1.
- [CGR 11] C. Cachin, R. Guerraoui, L. Rodrigues: *Introduction to Reliable and Secure Distributed Programming*. 2011. <https://distributedprogramming.net/>

In some chapters, further references are provided when selected topics will be discussed in greater detail.



Foundations of Distributed Systems in Context

Foundations of Distributed Systems is the mandatory introduction to the Master specialization in [Distributed Systems](#).

This course will be extended in Spring 2026 by specialized courses on *Computer Networks* (4 ECTS), *Distributed Information Systems* (6 ECTS), *High Performance Computing* (6 ECTS), *Privacy-Preserving Methods for Data Science and Distributed Systems* (6 ECTS) and a *Distributed Systems Project* (6 or 12 ECTS, individually via learning contract).

The lecture slides are on purpose more detailed than usual presentation slides. The objective is to come close to a script-like level of detail. However, these slides should not be considered as an alternative to a complete script and/or a textbook. They shall rather stimulate active participation and further reading.

Basel, September 2026

F. Ciorba
H. Schuldt
C. Tschudin
I. Wagner